Hints on Q2 and Q3 of Deep Learning Quiz

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It came to our attention that many students had trouble with Q2 and Q3 of Deep Learning Quiz. So we decided to provide some helpful hints.

*** Q2**: (paraphrasing a forum post posted 2 months ago)

For this one, you want to visualize a unit cube where each vertex represents a possible value of (x1,x2,x3). For instance, say we want a linear classifier for x1 AND x2 AND NOT x3.

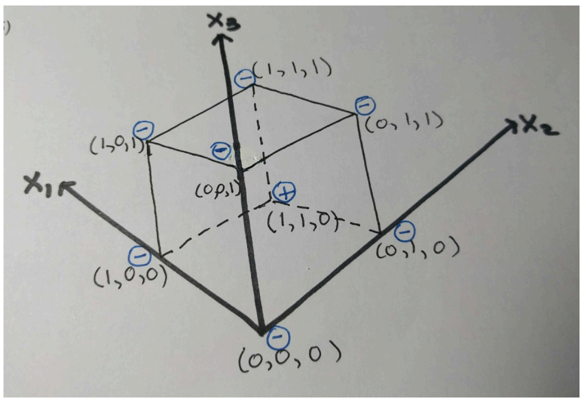
First we list all combinations of (x1,x2,x3) and corresponding outputs:

```
x1    x2    x3    | x1 AND x2 AND (NOT x3)
0     0     0     | 0
0     0     1     | 0
0     1     0     | 0
0     1     1     | 0
1     0     0     | 0
1     0     1     | 0
1     1     0     | 1
1     1     1     | 0
```

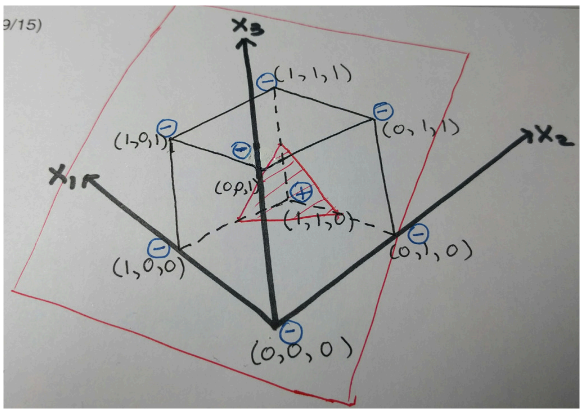
Now think of (x1,x2,x3) as coordinates in 3D space, and think of x1 AND x2 AND x3 as +/- label:

```
(x1,x2,x3) | label
(0,0,0)    | -
(0,0,1)    | -
(0,1,0)    | -
(0,1,1)    | -
(1,0,0)    | -
(1,0,1)    | -
(1,1,0)    | +
(1,1,1)    | -
```

Now visualize the 3D space with x1,x2,x3 axis. The 8 points would be vertices of a unit cube. Let us mark +/- near each vertex to indicate its label:



Now we ask: **"can we make a plane that separates (+)'s from (-)'s?"** That plane represents the linear classifier. It turns out we can:



This plane is obtained by connecting midpoints between (1,0,0), (1,1,0), and (0,1,0).

Repeat this analysis for all choices in Q2.

*** Q3:** First break down the Boolean expression into two subexpressions:

```
(x1 AND x2) OR (NOT x1 AND NOT x2) = z1 OR z2
```

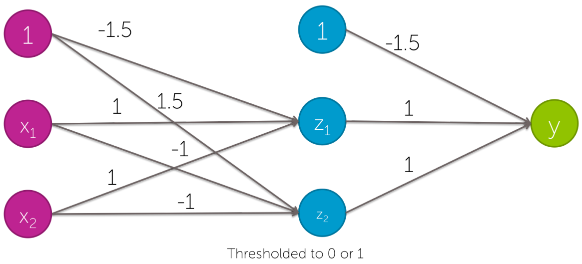
where

```
z1 = x1 AND x2
z2 = NOT x1 AND NOT x2
```

Then check two things:

- **Are z1 and z2 correctly implemented by the units in the middle layer?**
- **Does the last layer implement (z1 OR z2)?**

For instance, let us look at this two-layer neural network:



Truth table for z1:

```
x1    x2    z1      z1 thresholded    | (x1 AND x2)
0     0     -1.5    0                 | 0
0     1     -0.5    0                 | 0
1     0     -0.5    0                 | 0
1     1     0.5     1                 | 1
```

So yes, first unit in the middle layer correctly implements z1.

How about z2?

```
x1    x2    z2      z2 thresholded     | NOT x1 AND NOT x2
0     0     1.5     1                  | 1
0     1     0.5     1                  | 0
1     0     0.5     1                  | 0
1     1     -0.5    0                  | 0
```

No, second unit in the middle layer does not implement z2.

How about the second layer? Does it implement (z1 OR z2)?

```
z1    z2    y       y thresholded     | (z1 OR z2)
0     0     -1.5    0                 | 0
0     1     -0.5    0                 | 1
1     0     -0.5    0                 | 1
1     1     0.5     1                 | 1
```

So no, second layer does not implement (z1 OR z2).

Now repeat this line of reasoning for all other choices of Q3.

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